

Hua-Shan Shih / 施驊珊

Planetary Science | Atmospheric Escape | Dust-Ring Interactions

Personal Information

Current Location

- National Central University (NCU), Taiwan
(No. 300, Zhongda Rd., Zhongli District, Taoyuan City 32001, Taiwan)

E-mail

- hs.gaga.shih@gmail.com

Education

Ph.D. in Space Science and Engineering

- National Central University (NCU), Taiwan | Sep 2018 – Present
- Advisor: Prof. Wing-Huen Ip
- Dissertation (in progress): Atmospheric Evolution and Dust Dynamics in Solar System

M.A. in Space Science

- National Central University, Taiwan | Sep 2016 – June 2018
- Advisor: Prof. Wing-Huen Ip
- Thesis: *The Seasonal Evolution of the Atmosphere of Pluto and Other Dwarf Planets*

B.S. in Atmospheric Sciences

- National Central University, Taiwan | Sep 2011 – June 2015
- Advisors: Prof. Wing-Huen Ip & Prof. Tang-Huang Lin

Awards & Fellowship

Honorable Mention, Poster Competition – Physical Society of Taiwan (2025)

- *Photochemical Effects in Atmospheric Evolution of Early Mars*

Honorable Mention, Poster Competition – Physical Society of Taiwan (2022)

- *On Non-thermal Escape of Hot Oxygen Atoms from the atmosphere of Callisto*

Honorable Mention, Poster Competition – Physical Society of Taiwan (2020)

- *The Icing on Saturnian Rings – in the Past and Present*

Honorable Mention, Poster Competition – Physical Society of Taiwan (2019)

- *The Long-Term Evolution of Pluto's Seasonal Atmosphere*

Honorable Mention, Poster Competition – Physical Society of Taiwan (2018)

- *The Atmosphere of Pluto and Other Icy Dwarf Planets*

College Student Research Fellowship, National Science Council, Taiwan (2014 – 2015)

- Advisor: Prof. Wing-Huen Ip
- Project: *The Nucleus Model of C/2012 S1 (ISON)*

Research Experience

Graduate Research Assistant (Aug. 2016 – current)

- Institute of Astronomy, National Central University, Taiwan
- Advisor: Prof. Wing-Huen Ip
- Topic: *Particles dynamic on planetary atmosphere, thermal model of planets and satellites, ejecta from the impact event of small objects, etc.*

Visiting Scholar (Feb. 2024)

- Beihang University, Beijing, China
- Advisor: Prof. Haoyu Lyu
- Topic: *Mars' atmospheric escape in the presence of a 3D MHD model.*

Visiting Scholar (Summer in 2019)

- Southwest Research Institute, Boulder, United States
- Advisor: Dr. Leslie A. Young
- Topic: *Interaction between Pluto's Atmosphere and Surface*

Full-time Research Assistant (Mar. 2016 – July 2016)

- Institute of Space Science, National Central University, Taiwan
- PI: Prof. Chien-Ming Huang
- Project: *Investigation the Ionosphere during Magnetic Substorm through C/NOFS Data*

Full-time Research Assistant (July 2015 – Dec. 2015)

- Environmental Remote Sensing Laboratory, Center for Space and Remote Sensing Research, National Central University, Taiwan
- PI: Prof. Tang-Huang Lin
- Project: *Taiwan Integrated Research Program on Climate Change Adaptation Technology*

Programming Language

Matlab, C, Python

Teaching & Outreach

Guest Lectures & Public Talks

- **Solar Exploration Center**, Chiayi City | Oct 2025
[Topic TBD – Invited talk on Uranus]
Invited to present emerging research on Uranus and its mission to a public science audience.
- **Taipei Astronomical Museum**, Taipei City | Oct 2025
[Topic TBD]
Featured lecture for general audiences, bridging planetary science and public interest.
- **Taipei Astronomical Museum – General Education Series** | Mar 2025
The Possibility of Life on Mars: The Intersection of Science and Imagination
Explored astrobiological possibilities on Mars while contextualizing them in human culture and speculative thought.
- **Chiayi Astronomy Education Lecture Series**, Chiayi City | Nov 2024
Echoes of Orpheus: Navigating Mythology and Reality in the Cosmos
Delivered a talk fusing ancient mythology with modern space science to inspire public engagement.
- **Astronomy Carnival – Taiwan Space Union**, Taiwan | Jul 2024
The Martian Atmosphere Terminator: The Solar Wind's Conspiracy
Introduced the concept of atmospheric escape and terminator science using creative storytelling for outreach impact.
- **High School Astronomy Camp – Taipei Astronomical Museum**, Taipei | Oct 2022
My Research Odyssey: A Journey Through Planetary Science
Shared personal research experiences and career pathway to inspire students considering science education.
- **TTV Discovery Science – National Television Interview**, Taiwan | Jan 2024
The Mysteries of Saturn's Rings
Featured guest discussing the dynamics, structure, and origin theories of Saturn's ring system for a nationwide audience.

Teaching Assistantships

- **Department of Space Science and Engineering, NCU**
 - SS3026 – *Introduction to Lunar Science* (Spring 2021, Spring 2022)
 - SS3025 – *Introduction to Planetary Science* (Fall 2020, Fall 2021)
- **Department of Atmospheric Sciences, NCU**
 - AP3057 – *Introduction to Planetary Science I* (Fall 2019)
 - AP2052 – *Computer Programming and Graphics II* (Spring 2018)

Invited Talk

Astrophysics Lunch Seminar (Apr. 2022)

- National Center for Theoretical Science (NCTS), Taipei, Taiwan
- Topic: *Dynamics and formation of the Hydrogen Plume in the Vicinity of the Saturn's Ring*

Lunch Talk (Aug. 2018)

- Academia Sinica Institute of Astronomy and Astrophysics (ASIAA), Taipei, Taiwan
- Topic: *The Seasonal Evolution of the Atmospheres of Pluto and Other Dwarf Planets*

Conferences & Workshop

★ Oral Presentations

- **International Forum for Young Space Scientists**, Macau | Oct 2024
Study on Martian Hot Oxygen: Impact of Crustal Magnetic Fields in a 3D Global Model
- **American Geophysical Union (AGU) Fall Meeting**, San Francisco, USA | Dec 2023
Nano-dust Ring Rain at Uranus
- **Planetary Atmospheric Dynamics – From Mars, Titan to Pluto**, Taiwan | May 2019
The Long-term Evolution of Pluto's Atmosphere and Its Effect on Charon's Surface Tholin Formation
- **International Symposium on Lunar & Planetary Science**, Macao | Jun 2018
The Atmosphere of Pluto and Other Icy Dwarf Planets
- **Space Exploration and Scientific Observations with CubeSats**, Taiwan | Mar 2018
Volatile Transport on Mars
- **Climate Change in Taiwan for Adaptation Technology**, Taiwan | Sep 2015

📌 Poster Presentations

2025:

- Astronomical Society of the Republic of China 2025, Yunlin, Taiwan (May. 2025)
- Annual Meeting of the Physical Society of Taiwan 2025, Kaohsiung, Taiwan (Jan. 2025)

2024:

- Astronomical Society of the Republic of China 2024, Changhua, Taiwan (Jun. 2024)
- Annual Meeting of the Physical Society of Taiwan 2024, Taoyuan, Taiwan (Jan. 2024)

2023:

- Astronomical Society of the Republic of China 2023, Kaohsiung, Taiwan (May 2023)
- Annual Meeting of the Physical Society of Taiwan 2023, Tainan, Taiwan (Jan. 2023)

2022:

- Astronomical Society of the Republic of China 2022, Chiayi, Taiwan (Sep. 2022)
- The 19th Annual Meeting of the Asia Oceania Geosciences Society, Virtual (Aug. 2022)
- Annual Meeting of the Physical Society of Taiwan 2022, Tainan, Taiwan (Jan. 2022)

2021:

- The 18th Annual Meeting of the Asia Oceania Geosciences Society, Virtual (Aug. 2021)

2020:

- Astronomical Society of the Republic of China 2020, Taipei, Taiwan (Sep. 2020)
- Annual Meeting of the Physical Society of Taiwan 2020, Pingtung, Taiwan (Jan. 2020)

2019:

- The 16th Annual Meeting of the Asia Oceania Geosciences Society, Singapore (Aug. 2019)
- Pluto System After New Horizons, United States (July 2019)
- Astronomical Society of the Republic of China 2019, Taichung, Taiwan (May 2019)
- Annual Meeting of the Physical Society of Taiwan 2019, Hsinchu, Taiwan (Jan. 2019)

2018:

- Serendipities in the Solar System and Beyond, Taiwan (July 2018)
- Astronomical Society of the Republic of China 2018, Kinmen, Taiwan (May 2018)
- Annual Meeting of the Physical Society of Taiwan 2018, Taipei, Taiwan (Jan. 2018)

2017:

- Asian-Pacific Regional IAU Meeting 2017, Taipei, Taiwan (July 2017)
- Serendipities in the Solar System and Beyond, Taoyuan, Taiwan (July 2017)
- Astronomical Society of the Republic of China 2017, Miaoli, Taiwan (May 2017)
- Physical Society of the Republic of China 2017, Taipei, Taiwan (Jan. 2017)

2016:

- Physical Society of the Republic of China 2016, Kaohsiung, Taiwan (Jan. 2016)

Research Interests

- Planetary Atmospheres and Exospheres
- Non-thermal Escape and Photochemical Processes
- Dust Dynamics in Ring Systems
- Atmospheric Modeling and Seasonal Evolution of Outer Planetary Bodies

Publications

Peer-Reviewed Journal Articles

1. Hua-Shan Shih & Wing-Huen Ip (2023) *Infalling of nano-dust because of air drag on Uranus*. *Planetary and Space Science*, 237, 105782.
<https://doi.org/10.1016/j.pss.2023.105782>

Manuscripts in Preparation

1. Hua-Shan Shih, Wing-Huen Ip, Shibang Li, Haoyu Lu. *Global Martian Hot Oxygen Escape Influenced by Crustal Magnetic Fields* (in preparation).
2. Hua-Shan Shih, Ian-Lin Lai, Wing-Huen Ip. *Effects of Exogenic Dust on Uranus' Rings and Seasonal Variations* (in preparation).
3. Ian-Lin Lai, Hua-Shan Shih, Wei-Ling Tseng, Wing-Huen Ip. *Seasonal Variation of Dust Impacts on the Saturnian Rings* (in preparation).
4. Ian-Lin Lai, Hua-Shan Shih, Wei-Ling Tseng et al. *Water Transport from Enceladus to the Rings* (in preparation).